PARISHRAM 2.0 2026 (Class 12th)

PHYSICS

DPP: 02

Electric Charges and Fields

Q1 A positive charge +*q* is placed in a uniform electric field pointing to the right. Which direction will the particle accelerate?
(A) To the left (B) To the right
(C) Upward (D) Downward

Q2 The electric field due to a continuous charge distribution depends on:
(A) The geometry of the charge distribution and the location of the point of interest
(B) Only the total charge
(C) Only the distance from the charge
(D) None of the above

Q3 For a charge distribution with a volume charge density ρ, the total charge in a spherical volume of radius *R* is given by:
(A) $Q = \rho \cdot 4\pi R^2$
(B) $Q = \rho \cdot \frac{4}{3}\pi R^3$
(C) $Q = \rho \cdot \pi R^2$
(D) $Q = \rho \cdot \pi R^4$

Q4 **Assertion (A):** A charged particle in a uniform electric field experiences a constant force.
Reason (R): The force on a charged particle in an electric field is proportional to the charge and the field strength, and since the electric field is uniform, the force remains constant.
(A) Both Assertion and Reason are correct, and Reason is the correct explanation for Assertion.
(B) Both Assertion and Reason are correct, but Reason is not the correct explanation for Assertion.
(C) Assertion is correct, but Reason is incorrect.
(D) Assertion is incorrect, but Reason is correct.

Q5 The surface charge density σ is defined as:
(A) Charge per unit volume
(B) Charge per unit length
(C) Charge per unit area
(D) Charge per unit time

Q6 A charged particle with charge *q* and mass *m* is placed at rest in a uniform electric field *E*. What is the velocity of the particle after time *t*?
(A) $v = \frac{qE}{m}t$ (B) $v = \frac{m}{qE}t$
(C) $v = \frac{E}{m}t$ (D) $v = \frac{q}{mE}t$

Q7 A uniformly charged infinite line of charge with linear charge density λ creates an electric field at a point a distance *r* from the line. If the electric field at this point is *E*, what will happen to the electric field if the linear charge density λ is doubled?
(A) The electric field will increase by a factor of 2.
(B) The electric field will increase by a factor of 4.
(C) The electric field will decrease by a factor of 2.
(D) The electric field will remain unchanged.

Q8 If the linear charge density of a infinitely long cylinder is $4\mu\,\mathrm{Cm}^{-1}$ then electric field intensity at point 3.6 cm from axis is
(A) $4 \times 10^5\ \mathrm{NC}^{-1}$
(B) $2 \times 10^6\ \mathrm{NC}^{-1}$
(C) $8 \times 10^7\ \mathrm{NC}^{-1}$

(D) $12 \times 10^7\ \mathrm{NC}^{-1}$

Q9 What is the field at the centre of ring of radius *R* carrying a charge *q* uniformly distributed over it?

(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{3\sqrt{3}R^2}$

(B) Zero

(C) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{3\sqrt{3}R^2}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{3q}{2\sqrt{3}R^2}$

Q10 The electric field intensity due to a uniformly charged sphere is zero:

(A) at the centre

(B) at infinity

(C) at the centre and at infinite distance

(D) on the surface

Answer Key

Q1	B
Q2	A
Q3	B
Q4	A
Q5	C
Q6	A
Q7	A
Q8	B
Q9	B
Q10	C

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

A positive charge will experience a force in the direction of the electric field. Since the electric field is pointing to the right, the particle will accelerate to the right.

Video Solution:

Q2 Text Solution:

The electric field at a point due to a continuous charge distribution depends on the geometry of the distribution and the location of the point.

Video Solution:

Q3 Text Solution:

The total charge in a spherical volume with radius R and volume charge density ρ is given by the volume of the sphere $\frac{4}{3}\pi R^3$ multiplied by ρ.

Thus, $Q = \rho \cdot \frac{4}{3}\pi R^3$

Video Solution:

Q4 Text Solution:

Both Assertion and Reason are correct, and Reason is the correct explanation for Assertion.

Video Solution:

Q5 Text Solution:

Surface charge density σ is the charge per unit area, typically expressed as $\sigma = \frac{Q}{A}$, where Q is the charge and A is the surface area.

Video Solution:

Q6 Text Solution:

The acceleration is $a = \frac{qE}{m}$. Using $v = v_0 + at$ with $v_0 = 0$, we get $v = \frac{qE}{m}t$.

Video Solution:

Q7 Text Solution:

The electric field due to a uniformly charged infinite line is given by $E = \frac{2k\lambda}{r}$. If λ is doubled, the electric field will also double.

Video Solution:

Q8 **Text Solution:**

For a charged cylinder

$$E = \frac{\lambda}{2\pi\varepsilon_0 r} = \frac{4\times10^{-6}\times18\times10^{9}}{3.6\times10^{-2}}$$

$$= 2 \times 10^{6}\ \mathrm{NC}^{-1}$$

Video Solution:

Q9 **Text Solution:**

Electric field at the centre of ring due to charge q distributed over it is zero.

Video Solution:

Q10 **Text Solution:**

$$E_{in} = \frac{\rho r}{3\varepsilon_0}$$

At r = 0 (centre)

E = 0

At infinite distance sphere behave like a point charge

$E = \frac{Kq}{r^2}, r = \infty$, so E = 0

Video Solution:

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